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Report

On

IPL Teams Analytics Dashboard

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Data Analysis and Visualization (3151608)

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1. Introduction

In today's digital era, data analysis and visualization play an important role in understanding large datasets and extracting meaningful insights. With the increasing amount of data generated in sports analytics, it becomes necessary to use advanced tools to analyze performance effectively.

This project focuses on the analysis of IPL (Indian Premier League) cricket data using Power BI. IPL is one of the most popular cricket leagues, generating large amounts of match data including runs, wickets, overs, and team performance. Analyzing this raw data manually is difficult and time-consuming.

The main objective of this project is to develop interactive dashboards that provide clear insights into batting and bowling performance. Two dashboards are created:

- Batting Analysis Dashboard
- Bowling & Wicket Analysis Dashboard

These dashboards help users understand scoring patterns, team performance, and match trends in an easy and visual way.

2. Problem Statement

IPL datasets contain large volumes of data, making it difficult to analyze performance using traditional methods like Excel sheets.

The main problems are:

- Difficulty in understanding raw cricket data
- Lack of visual representation
- Time-consuming manual analysis
- Difficulty in comparing team performance

This project aims to answer the following questions:

- Which team scores the highest runs?
- How do runs increase over overs?
- Which team takes the most wickets?
- How do wickets affect match outcomes?
- What is the importance of last 5 overs?

By solving these problems, the project provides better insights using visualization tools.

3. Dataset Description

The dataset contains match-level information of IPL cricket matches. It includes both batting and bowling performance attributes such as runs, wickets, overs, and team details. The dataset is useful for analyzing scoring trends, team performance, and match dynamics using visualization tools like Power BI.

Source: Retail Orders Dataset (cleaned), used for academic analysis under the subject Data Analysis and Visualization (3151608), B.E. Semester VI, Information Technology Department, C.
K. Pithawalla College of Engineering and Technology.

The following table provides a detailed description of each column in the dataset:

No.	Column Name	Data Type	Description
1	bat_team	String	Name of the batting team
2	bowl_team	String	Name of the bowling team
3	overs	Float	Number of overs completed
4	runs	Integer	Total runs scored by batting team
5	wickets	Integer	Number of wickets lost
6	runs_last_5	Integer	Runs scored in last 5 overs
7	wickets_last_5	Integer	Wickets lost in last 5 overs
8	total	Integer	Final total score of the innings
9	Runrate	Float	Run rate calculated as runs divided by overs

4. Data Preprocessing

Data cleaning is the most vital phase of the project. To ensure our report is based on high-quality data, we implemented a multi-stage cleaning pipeline:

4.1 Consistent Team Filtering:

We identified and retained only the eight "consistent" franchises (CSK, MI, KKR, RCB, KXIP, RR, DD, and SRH). This ensured that our analysis was based on a significant sample size for each team. This step filtered out over 22,000 rows of irrelevant data.

######	Kolkata Knight Riders	Royal Challengers Bangalore	185	3	18.1	60	1	222	10.22
######	Kolkata Knight Riders	Royal Challengers Bangalore	185	3	18.2	59	1	222	10.16
######	Kolkata Knight Riders	Royal Challengers Bangalore	191	3	18.3	64	1	222	10.43
######	Kolkata Knight Riders	Royal Challengers Bangalore	195	3	18.4	67	1	222	10.59
######	Kolkata Knight Riders	Royal Challengers Bangalore	196	3	18.5	66	1	222	10.59
######	Kolkata Knight Riders	Royal Challengers Bangalore	200	3	18.6	66	1	222	10.75
######	Chennai Super Kings	Kings XI Punjab	46	1	5.1	45	1	240	9.019
######	Chennai Super Kings	Kings XI Punjab	52	1	5.2	51	1	240	
######	Chennai Super Kings	Kings XI Punjab	52	1	5.3	47	1	240	9.811
04-2008	Chennai Super Kings	Kings XI Punjab	53	1	5.4	48	1	240	9.814
######	Chennai Super Kings	Kings XI Punjab	53	1	5.5	44	1	240	9.636
######	Chennai Super Kings	Kings XI Punjab	53	1	5.6	40	1	240	9.464
######	Chennai Super Kings	Kings XI Punjab	57	1	6.1	40	1	240	9.344
######	Chennai Super Kings	Kings XI Punjab	57	2	6.2	38	2	240	9.193
######	Chennai Super Kings	Kings XI Punjab	58	2	6.3	38	2	240	9.206
######	Chennai Super Kings	Kings XI Punjab	60	2	6.4	40	2	240	9.

4.2 Phase-Based Truncation:

To ensure our dashboard reflects the true "build-up" of an innings, we removed all data before the **5.0 over** mark. By doing so, we focus the analysis on how teams manage their wickets and run rates during the crucial middle and death overs.

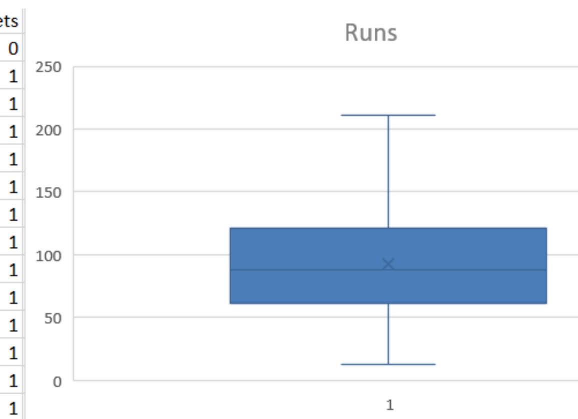
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	0	5.1	59	0	222	11.96078
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	1	5.2	59	1	222	11.73077
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	1	5.3	59	1	222	11.50943
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	1	5.4	59	1	222	11.2963
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	1	5.5	58	1	222	11.09091
########	Kolkata Knight Riders	Royal Challengers Bangalore	61	1	5.6	58	1	222	10.89286
########	Kolkata Knight Riders	Royal Challengers Bangalore	62	1	6.1	55	1	222	10.16393
##18-04-2008	Kolkata Knight Riders	Royal Challengers Bangalore	63	1	6.2	52	1	222	10.16129
########	Kolkata Knight Riders	Royal Challengers Bangalore	64	1	6.3	47	1	222	10.15873
########	Kolkata Knight Riders	Royal Challengers Bangalore	66	1	6.4	45	1	222	10.3125
########	Kolkata Knight Riders	Royal Challengers Bangalore	67	1	6.5	46	1	222	10.30769
########	Kolkata Knight Riders	Royal Challengers Bangalore	68	1	6.6	47	1	222	10.30303

4.3 Statistical Outlier Removal (IQR):

We applied the Interquartile Range (IQR) method to the total score.

- **Q1 (142 runs)** and **Q3 (183 runs)** were calculated.
- The **Lower Bound** was set at **80.5**. Any match where a team scored less than 81 runs was discarded as an outlier (typically rain-curtailed), leaving us with **39,728 clean records** for analysis.

bat_team	bowl_team	runs	wickets
Kolkata Knight Riders	Royal Challengers Bangalore	61	0
Kolkata Knight Riders	Royal Challengers Bangalore	61	1
Kolkata Knight Riders	Royal Challengers Bangalore	61	1
Kolkata Knight Riders	Royal Challengers Bangalore	61	1
Kolkata Knight Riders	Royal Challengers Bangalore	61	1
Kolkata Knight Riders	Royal Challengers Bangalore	61	1
Kolkata Knight Riders	Royal Challengers Bangalore	62	1
Kolkata Knight Riders	Royal Challengers Bangalore	63	1
Kolkata Knight Riders	Royal Challengers Bangalore	64	1
Kolkata Knight Riders	Royal Challengers Bangalore	66	1
Kolkata Knight Riders	Royal Challengers Bangalore	67	1
Kolkata Knight Riders	Royal Challengers Bangalore	68	1
Kolkata Knight Riders	Royal Challengers Bangalore	68	1
Kolkata Knight Riders	Royal Challengers Bangalore	69	1



4.4 Feature Engineering

Feature engineering is an important step in data preprocessing where new useful features are created from existing data to improve analysis. In this project, several derived features were created to better understand the performance of teams.

A new column called **Run Rate** was calculated using the formula:

Run Rate = Rate / Overs

runs	wickets	overs	runs_last_5	wickets_last_5	total	Runrate
61	0	5.1	59	0	222	11.96078
61	1	5.2	59	1	222	11.73077
61	1	5.3	59	1	222	11.50943
61	1	5.4	59	1	222	11.2963
61	1	5.5	58	1	222	11.09091
61	1	5.6	58	1	222	10.89286
62	1	6.1	55	1	222	10.16393
63	1	6.2	52	1	222	10.16129
64	1	6.3	47	1	222	10.15873
66	1	6.4	45	1	222	10.3125
67	1	6.5	46	1	222	10.30769
68	1	6.6	47	1	222	10.30303

5. Methodology

This project adopted a structured, phased methodology to analyze the retail sales dataset and develop an interactive Power BI dashboard. The following steps describe the complete process:

➔ Tools and Technologies Used

- Microsoft Excel (.xlsx): Used as the primary data storage format for the cleaned dataset.
- Microsoft Power BI Desktop: Used as the primary tool for building interactive, multi-page analytical dashboards with KPI cards, bar charts, donut charts, scatter plots, map visuals, and line/combo charts.

➔ Step-wise Process

- **Step 1 – Data Loading:** Dataset loaded into Power BI
- **Step 2 – Data Verification:** Checked data types and errors using Power Query
- **Step 3 – Measure Creation:** Created KPIs using DAX (Sales, Profit, Orders, etc.)
- **Step 4 – Dashboard Design:** Built 5-page dashboard with different analysis views
- **Step 5 – Chart Selection:** Used suitable charts (bar, line, donut, map, scatter)
- **Step 6 – Interactivity:** Enabled filters and navigation between pages
- **Step 7 – Validation:** Verified all KPI values for accuracy

6. Analysis & Visualization

The dataset was analyzed using various visualizations in Power BI to understand team performance, scoring patterns, and wicket distribution. The following key analyses were performed:

6.1 Total Runs Analysis

The total runs scored by each team were analyzed to compare batting performance. Mumbai Indians have the highest runs (545,396), followed by Chennai Super Kings (507,782) and Kings XI Punjab (497,139), indicating strong batting performance. Sunrisers Hyderabad have the lowest total runs (317,910), showing comparatively lower performance.

Row Labels	Sum of runs
Chennai Super Kings	507782
Delhi Daredevils	375571
Kings XI Punjab	497139
Kolkata Knight Riders	401595
Mumbai Indians	545396
Rajasthan Royals	399040
Royal Challengers Bangalore	422735
Sunrisers Hyderabad	317910
Grand Total	**3467168**

6.2 Total Wickets Analysis

The total wickets across all IPL matches are approximately above 100K, indicating strong bowling involvement. Teams like Delhi Daredevils and Kolkata Knight Riders have taken the highest number of wickets. This shows that higher wicket-taking ability helps teams control the match effectively.

Row Labels	Sum of wickets
Chennai Super Kings	12385
Delhi Daredevils	16656
Kings XI Punjab	12019
Kolkata Knight Riders	16501
Mumbai Indians	14355
Rajasthan Royals	13651
Royal Challengers Bangalore	15069
Sunrisers Hyderabad	5259
Grand Total	**105895**

6.3 Runs by Team

The average total score for each IPL team was analyzed to understand batting performance consistency. The analysis shows how different teams perform on average in terms of total runs scored.

Row Labels	Average of total
Chennai Super Kings	161.8803884
Delhi Daredevils	162.5885226
Kings XI Punjab	170.2069115
Kolkata Knight Riders	157.9708016
Mumbai Indians	164.9668849
Rajasthan Royals	158.2977324
Royal Challengers Bangalore	164.8149512
Sunrisers Hyderabad	163.6420455
Grand Total	**162.9605936**

6.4 Average Run Rate Analysis

The average run rate of each IPL team was analyzed to understand the scoring efficiency of teams. Run rate represents the number of runs scored per over and is an important metric to evaluate batting performance.

Row Labels	Average of Runrate
Chennai Super Kings	7.61188034
Delhi Daredevils	7.537640343
Kings XI Punjab	7.963554566
Kolkata Knight Riders	7.588692409
Mumbai Indians	7.711154853
Rajasthan Royals	7.294990414
Royal Challengers Bangalore	7.873160533
Sunrisers Hyderabad	7.530933345
Grand Total	**7.650907297**

6.5 Last 5 Overs Analysis

The last 5 overs contribute significantly to both runs and wickets in a match. Teams score aggressively during this phase, which also increases the chances of losing wickets. This makes the final overs crucial for deciding the match outcome.

Row Labels	Average of runs_last_5
Chennai Super Kings	38.54618664
Delhi Daredevils	38.37083551
Kings XI Punjab	40.42621169
Kolkata Knight Riders	37.54415148
Mumbai Indians	39.44284378
Rajasthan Royals	36.98344671
Royal Challengers Bangalore	39.56600994
Sunrisers Hyderabad	39.00774793
Grand Total	**38.71536875**

7. Dashboard

The dashboard is developed using Power BI and includes two pages: IPL Batting Analysis and IPL Bowling Analysis. It displays key KPIs like average score, total runs, total wickets, and last 5 overs performance using interactive charts and filters.

7.1 IPL Batting Analysis Dashboard

The IPL Batting Analysis dashboard provides a comprehensive overview of batting performance across different teams. It includes key performance indicators such as **Average Score (162.96)**, **Maximum Score (240)**, and **Total Runs (3M)**, which give a quick summary of overall batting performance.

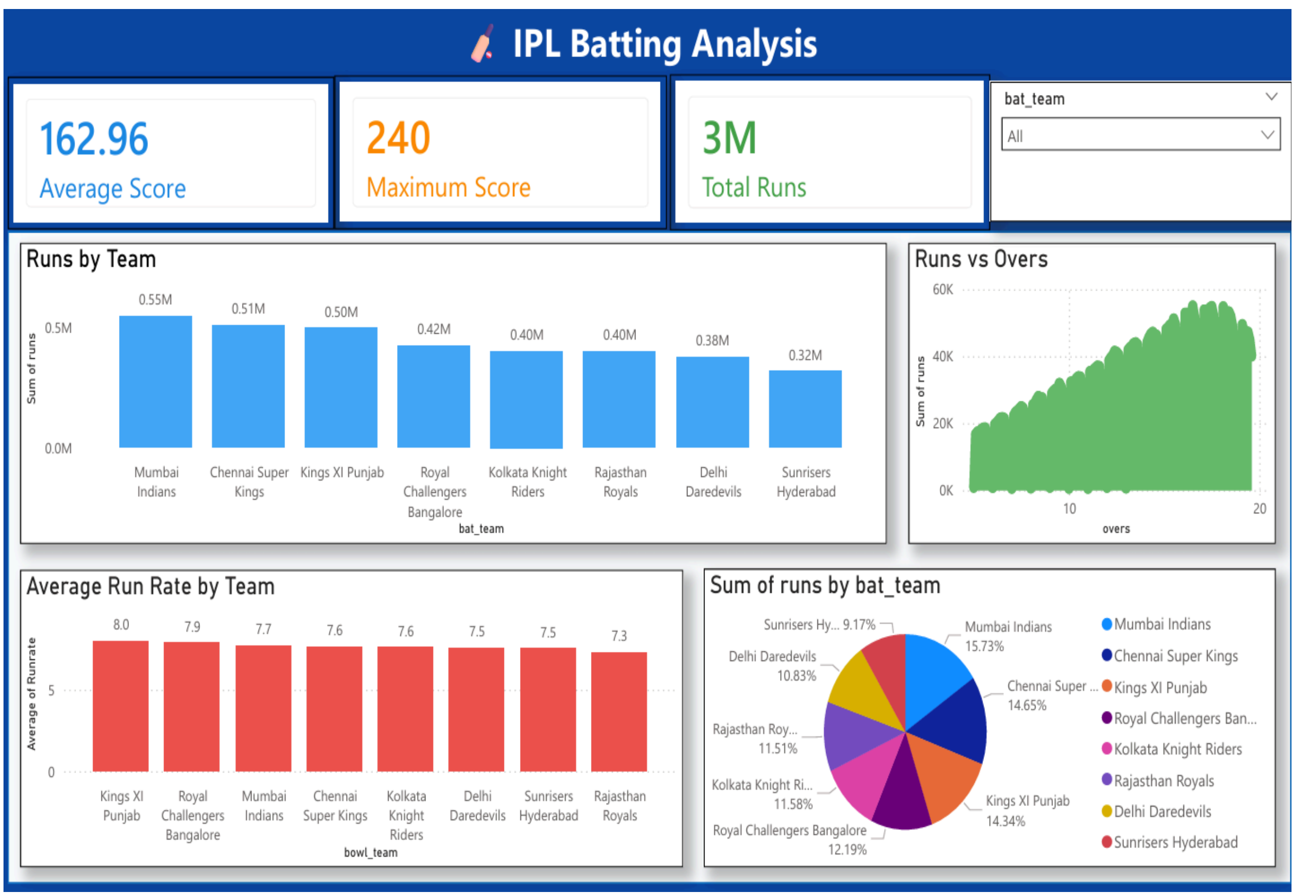


Figure 7.1: IPL Batting Analysis Dashboard

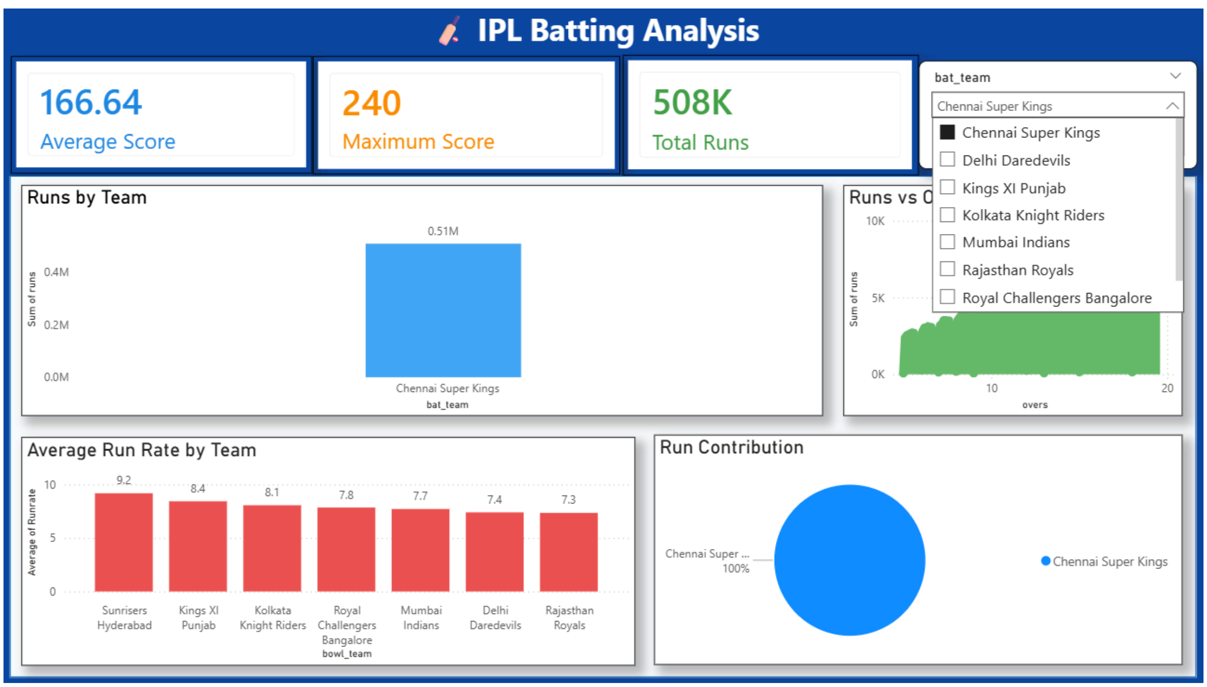
The **Runs by Team** bar chart shows that Mumbai Indians have the highest total runs

(~0.55M), followed by Chennai Super Kings and Kings XI Punjab, indicating strong batting consistency. The **Runs vs Overs** line chart illustrates that runs increase steadily as overs progress, with rapid growth in the later overs, highlighting aggressive scoring towards the end of the innings.

The **Average Run Rate by Team** chart compares scoring efficiency, showing that teams like Kings XI Punjab and Royal Challengers Bangalore have higher run rates. The **Team Contribution pie chart** displays the percentage contribution of each team to total runs, helping in understanding the distribution of scoring among teams.

Additionally, the dashboard includes a **batting team filter**, allowing users to select specific teams and analyze their performance dynamically.

7.1.1IPL Team-Specific Batting Analysis Dashboard



This dashboard provides a detailed analysis of a selected IPL team's batting performance using interactive filters. It displays key metrics such as average score (166.64), maximum score (240), and total runs (508K) for Chennai Super Kings. The visualizations show run trends across overs, run rate comparison, and team contribution, helping in understanding performance patterns. This allows users to analyze individual team performance more effectively.

7.2 IPL Bowling Analysis Dashboard

The IPL Bowling Analysis dashboard focuses on wicket-taking performance and bowling effectiveness. It includes important KPIs such as **Total Wickets (106K)**, **Average Wickets (2.84)**, and **Wickets in Last 5 Overs (45K)**, which summarize overall bowling performance.

The **Wickets by Bowling Team** bar chart shows that teams like Delhi Daredevils and Kolkata Knight Riders have taken the highest number of wickets, indicating strong bowling performance. The **Wickets vs Overs** chart illustrates that wickets increase as the match progresses, with the highest number of wickets occurring in the later overs due to increased pressure.

The **Wickets in Last 5 Overs** chart highlights the importance of the final overs, showing that a significant number of wickets are taken during this phase. The **Wicket Contribution pie chart** represents the percentage distribution of wickets among teams, helping to identify major contributors in bowling.

A **bowling team filter** is also included, enabling users to analyze specific team performance interactively.

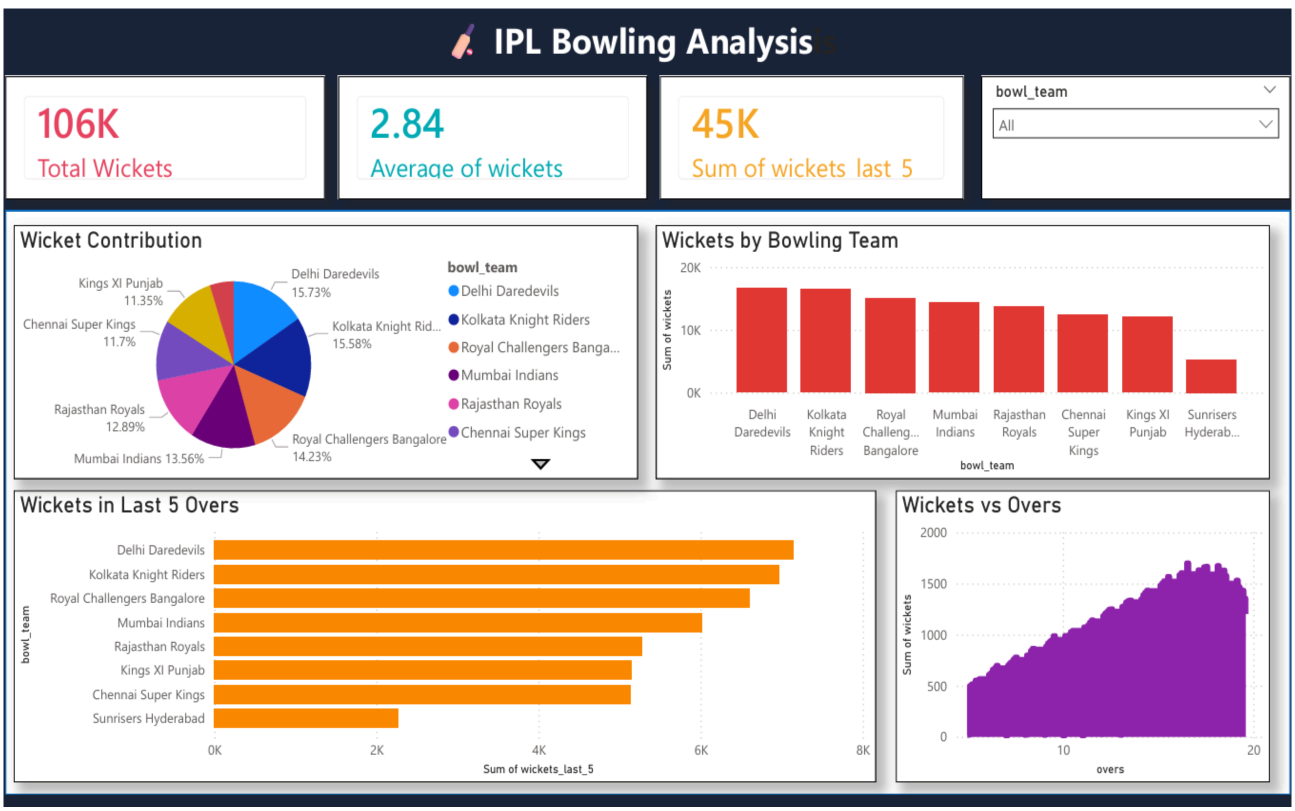


Figure 7.2: IPL Bowling Analysis Dashboard

8. Interpretation & Insights

The following key patterns and insights were identified through the analysis and visualization of the IPL dataset:

8.1 Strong Batting Performance by Top Teams

Teams like Mumbai Indians, Chennai Super Kings, and Kings XI Punjab contribute the highest runs in the dataset. These teams show consistent batting performance with higher average scores and run rates. This indicates strong batting lineups and better scoring ability.

8.2 Run Rate Reflects Scoring Efficiency

Run rate analysis shows that teams with higher run rates, such as Kings XI Punjab and Royal Challengers Bangalore, perform more efficiently in scoring. Higher run rate directly impacts total runs and improves match performance.

8.3 Wickets Play a Key Role in Match Control

Teams like Delhi Daredevils and Kolkata Knight Riders take the highest number of wickets. Higher wicket-taking ability helps in restricting the opponent's score and increases chances of winning matches.

8.4 Importance of Last 5 Overs

The last 5 overs contribute significantly to both runs and wickets. Teams score aggressively during this phase, but also lose more wickets due to risky shots. This makes the final overs the most crucial part of the match.

9. Conclusion

This project successfully demonstrates the application of data analysis and visualization techniques on an IPL cricket dataset. By using Microsoft Power BI for interactive dashboard development and Microsoft Excel for data preprocessing, a comprehensive analytical solution was created. The project includes two dashboards: IPL Batting Analysis and IPL Bowling Analysis, which provide insights into team performance, scoring patterns, and wicket distribution.

The analysis of the dataset revealed that teams such as Mumbai Indians, Chennai Super Kings, and Kings XI Punjab contribute significantly to total runs, indicating strong batting performance. The overall average score of around 163 runs reflects a competitive scoring trend in IPL matches. Additionally, teams like Delhi Daredevils and Kolkata Knight Riders show strong bowling performance with higher wicket counts, highlighting their ability to control matches.

Key observations include the importance of run rate in determining scoring efficiency and the significant impact of the last 5 overs on match outcomes. It was observed that both runs and wickets increase during the final overs, making this phase crucial for both batting and bowling teams. The analysis also highlights that balanced performance in both batting and bowling is essential for overall success.

The Power BI dashboards developed in this project serve as an interactive and effective tool for analyzing cricket data. These dashboards allow users to explore data dynamically using filters and visualizations, making it easier to identify trends and patterns. This project emphasizes the importance of data analytics in sports and demonstrates how visualization can simplify complex datasets.

10. Future Scope

While this project provides a comprehensive analysis of the IPL dataset, several improvements and enhancements can be made in the future:

- **Real-Time Data Integration:** The current analysis is based on static data. Future work can include real-time IPL match data using APIs, allowing live dashboard updates and real-time performance monitoring.
- **Player-Level Analysis:** Future enhancements can include detailed player performance analysis such as individual runs, strike rate, wickets, and economy rate to gain deeper insights.
- **Predictive Analytics:** Machine learning models can be applied to predict match outcomes, team performance, and winning probabilities based on historical data.
- **Advanced Visualizations:** Additional advanced charts and interactive visuals can be added to improve dashboard usability and provide more detailed insights.
- **Mobile Dashboard Optimization:** The dashboard can be optimized for mobile devices using Power BI mobile features, enabling users to access insights anytime and anywhere.
- **Team Strategy Analysis:** The dashboard can be extended to analyze team strategies such as batting order, powerplay performance, and death overs strategy.
- **Integration with Other Leagues:** The model can be expanded to include other cricket leagues like T20 World Cup or Big Bash League for comparative analysis.

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